

Analysis

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2026-05-25

Analysis

Packages

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.4
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

Functions Script

```
source('/Users/camsmithers/Desktop/Camalytics/NBA/inR/Functions.R')
```

Data Sets

Loading in the data sets that I'll need for analysis.

```
team_box_scores <- readRDS(
  '/Users/camsmithers/Desktop/Camalytics/NBA/Data-NBA/Current/team_box_scores_updated.rds')
player_box_scores <- readRDS(
  '/Users/camsmithers/Desktop/Camalytics/NBA/Data-NBA/Current/full_box_player_updated.rds')
team_season_stats <- readRDS(
  '/Users/camsmithers/Desktop/Camalytics/NBA/Data-NBA/Current/team_season_stats_updated.rds')
player_season_stats <- readRDS(
  '/Users/camsmithers/Desktop/Camalytics/NBA/Data-NBA/Current/player_season_stats_updated.rds')

write_csv(team_box_scores,
  '/Users/camsmithers/Desktop/Camalytics/NBA/Data-NBA/Current/tbs.csv')
write_csv(player_box_scores,
  '/Users/camsmithers/Desktop/Camalytics/NBA/Data-NBA/Current/pbs.csv')
write_csv(team_season_stats,
  '/Users/camsmithers/Desktop/Camalytics/NBA/Data-NBA/Current/tss.csv')
write_csv(player_season_stats,
  '/Users/camsmithers/Desktop/Camalytics/NBA/Data-NBA/Current/pss.csv')
```

Applying a function I wrote that fixes some aesthetic issues with player names. This will occur with foreign players that have special characters or foreign characters in their name.

```
player_box_scores <- fixplayername(player_box_scores)
player_season_stats <- fixplayername(player_season_stats)
```

Outcomes and Small Metrics

Outcome I'm adding the points from the opposing team so I can create an outcome variable based on the margin of the outcome. This will be multinomial regression with four different classifications. Those are...

```
opponent_stats <- team_box_scores %>%
  select(gamedate, team, pts) %>%
  rename('opp_pts'='pts')

team_box_scores <- team_box_scores %>%
  left_join(opponent_stats, by = c('gamedate', 'opponent'='team'))

team_box_scores <- team_box_scores %>%
  mutate(
    ptsdiff = pts - opp_pts,
    twoleveloc = ifelse(offrating > defrating, 1, 0),
    threeleveloc = case_when(
      ptsdiff >= 11 ~ 2,
      ptsdiff %in% 1:10 ~ 1,
      TRUE ~ 0),
    fourleveloc = case_when(
      ptsdiff <= -11 ~ 0,
      ptsdiff %in% -1:-10 ~ 1,
      ptsdiff %in% 1:10 ~ 2,
      ptsdiff >= 11 ~ 3)) %>%
  mutate(
    twoleveloc = factor(twoleveloc,
                        levels = c(0, 1),
                        labels = c('Loss', 'Win')),
    threeleveloc = factor(threeleveloc,
                           levels = c(0, 1, 2),
                           labels = c('Loss', 'Basic Win', 'Blowout Win')),
    fourleveloc = factor(fourleveloc,
                           levels = c(0, 1, 2, 3),
                           labels = c('Blowout Loss', 'Basic Loss', 'Basic Win', 'Blowout Win')))
```

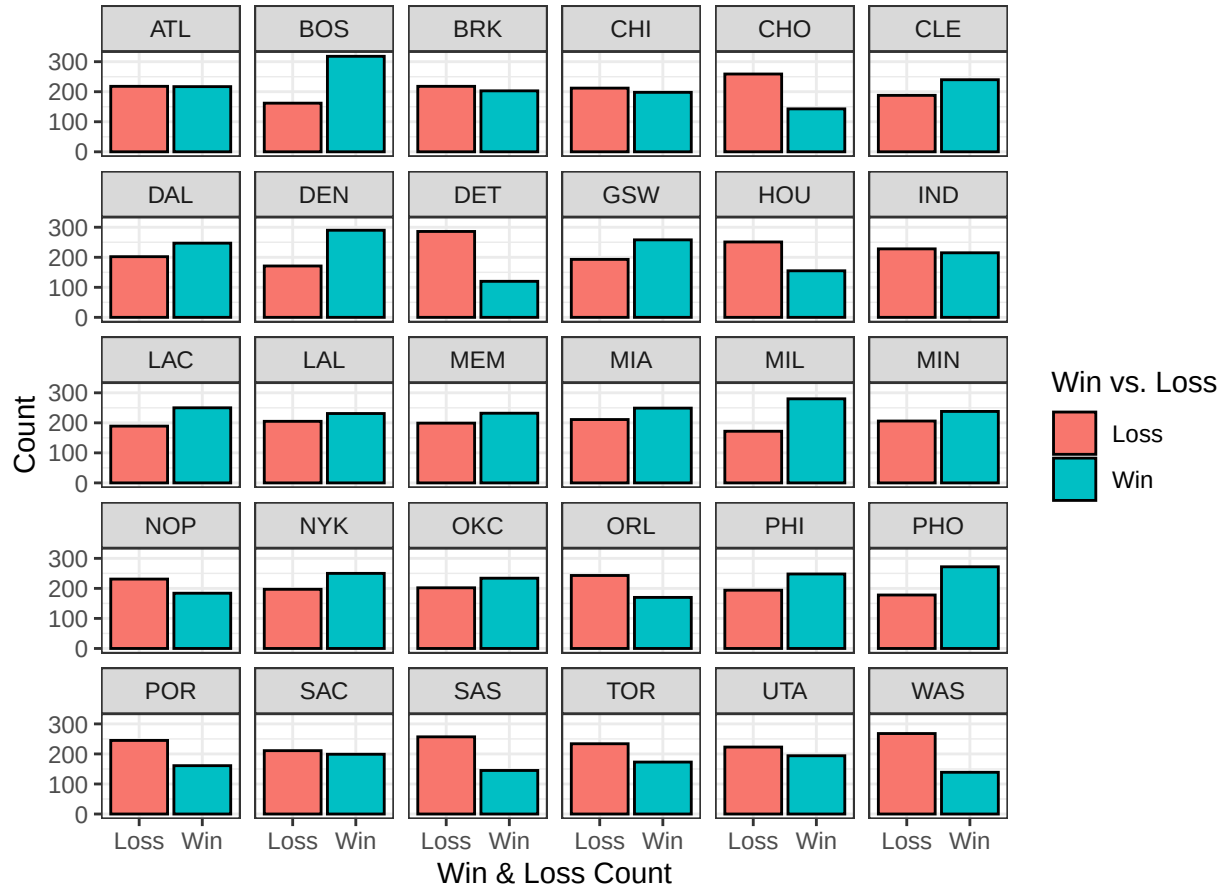
Outcome Visualizations

```
plot1 <- ggplot(team_box_scores,
                aes(x=twoleveloc,
                    fill=twoleveloc)) +
  geom_bar(position='dodge', color='black') +
  labs(
    title='Wins vs. Losses from 2021 to 2025',
    x='Win & Loss Count',
    y='Count',
    fill='Win vs. Loss') +
  facet_wrap(~team) +
```

```
theme_bw()
plot1
```

Plot 1: Wins vs. Losses

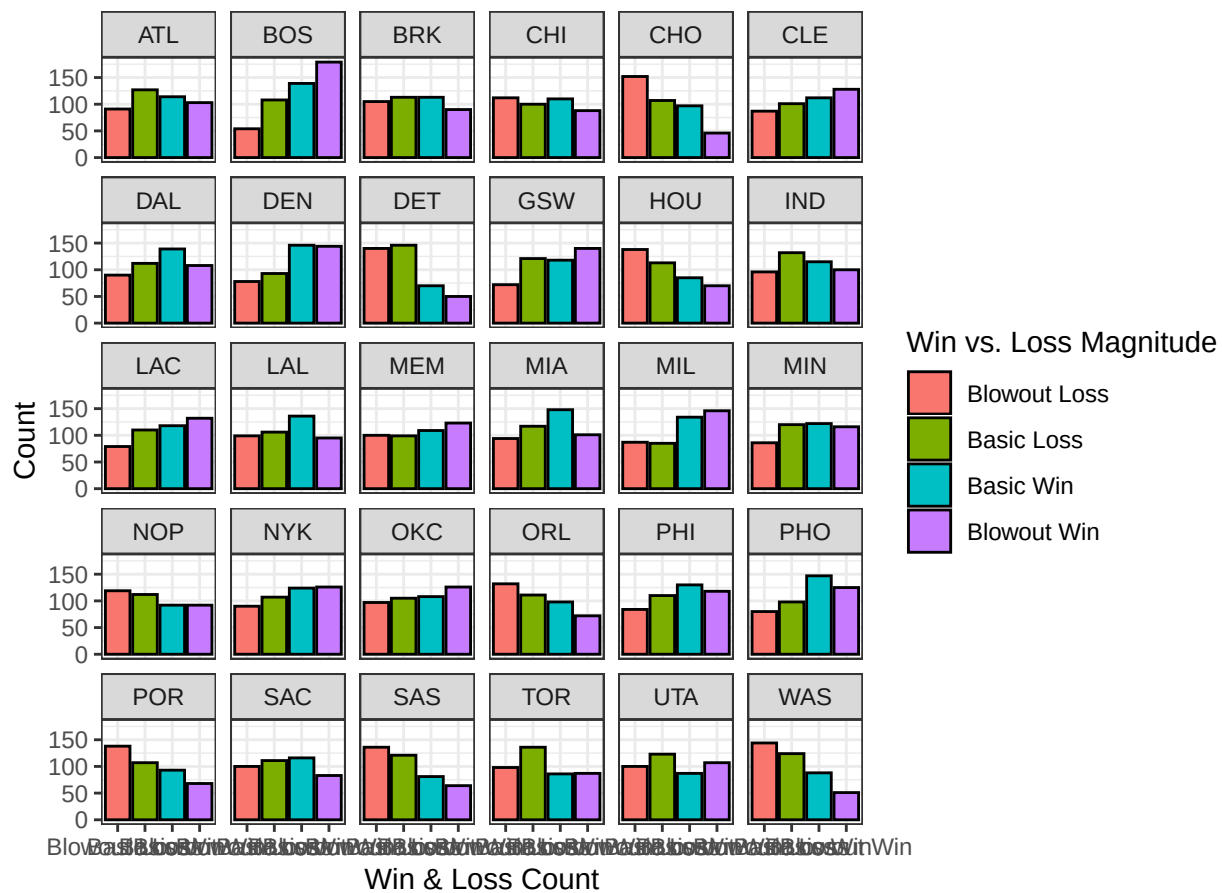
Wins vs. Losses from 2021 to 2025



```
plot2 <- ggplot(team_box_scores,
  aes(x=fourlevelloc,
    fill=fourlevelloc)) +
  geom_bar(position='dodge', color='black') +
  labs(
    title='Magnitude of Wins vs. Losses from 2021 to 2025',
    x='Win & Loss Count',
    y='Count',
    fill='Win vs. Loss Magnitude') +
  facet_wrap(~team) +
  theme_bw()
plot2
```

Plot 2: 4 Level Outcome Variable

Magnitude of Wins vs. Losses from 2021 to 2025



I want to look at how missing a player affects the chance of a team to win a game; however, I have to determine which players are important for the outcome of the game. I'm going to look at few metrics and use those to determine a player's significance. For the different metrics, different players may end up being the most important player. Those metrics include...

1. Win Shares: `ws`
2. Value Over Replacement Player (VORP): `vorp`
3. Player Efficiency Rating: `player_effrtg`

```
bp_by_ws <- best_player_metric(player_season_stats, ws, 20, 10)
bp_by_vorp <- best_player_metric(player_season_stats, vorp, 20, 10)
bp_by_per <- best_player_metric(player_season_stats, player_effrtg, 20, 10)
```

Joining the data from the best player by a certain metric data sets to later determine if player was on the court and how that affected the outcome.

```
team_box_scores_2 <- team_box_scores %>%
  left_join(bp_by_ws, by = c('team', 'season'='year')) %>%
  left_join(bp_by_vorp, by = c('team', 'season'='year', 'name')) %>%
  left_join(bp_by_per, by = c('team', 'season'='year', 'name'))
```

Things needed to create this conditional outcome variable or variables...